

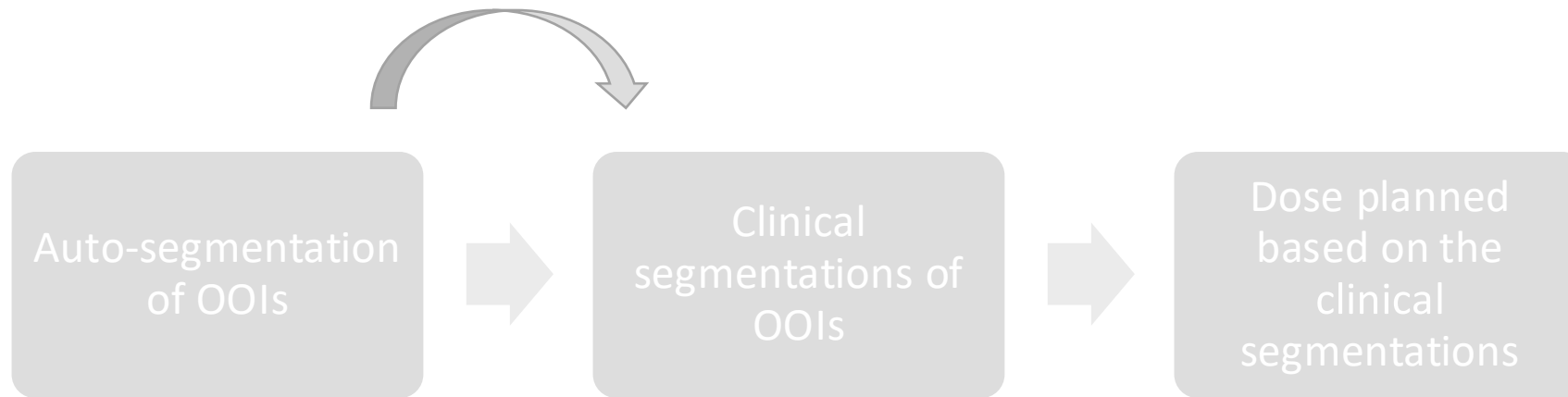
Recognizing dosimetrically irrelevant edits of auto-segmentations in H&N OOLs

MIART – 01/10/2026

Mar Fernandez Salamanca

What is happening right now in the clinic?

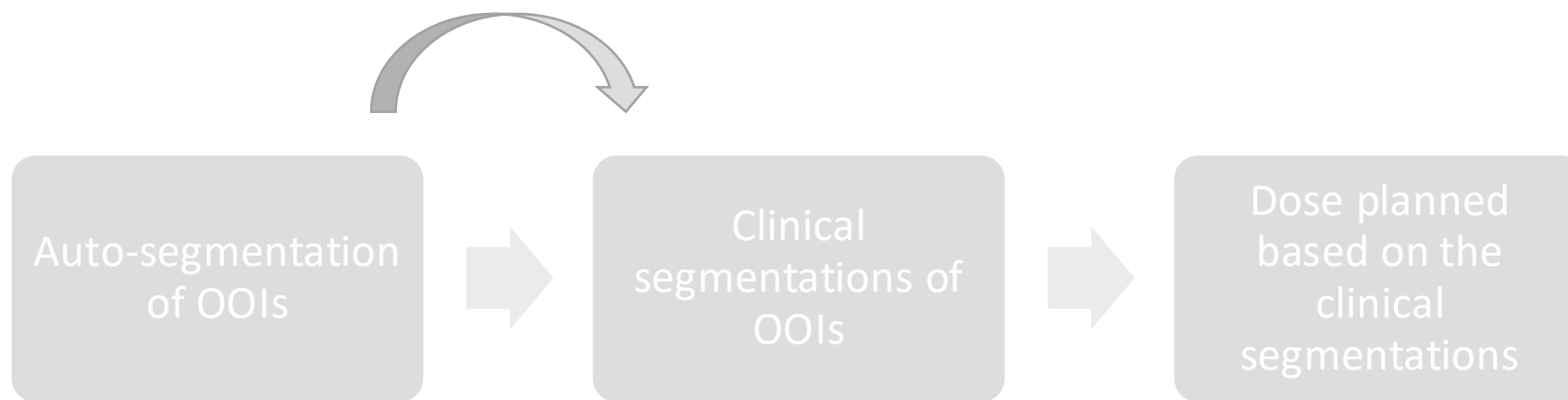
RTTs correct the OOI's auto-segmentation
to be clinically acceptable



- Manual editing of auto-seg of OOI's remains the bottleneck in RT
- The lack of clear criteria leads to under- or over-editing without knowledge on dosimetric effect
- Previous work has shown that the spatial relationship between OOI's and planning target volume is linked to dosimetric impact.

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- NKI timeline of different auto-seg models for H&N OOI's:
 - January 2018 → Atlas based (11 atlases, Workflowbox Mirada)
 - July 2021 → Atlas based (22 atlases, Workflowbox Mirada)
 - January 2023 → Deep learning (DLCExpert, Mirada)
 - May 2026 → Deep learning (MIM, Elekta)

AIM (1): Assess the relationship between manual editing patterns and their dosimetric impact

AIM (2): Determine the relationship between OOI-PTV distance and dosimetric impact, and translate it into editing recommendations

AIM (1): Assess the relationship between manual editing patterns and their dosimetric impact

How can we measure the dosimetric impact of those edits?

Ideally we would calculate the DIE (dosimetric impact on evaluation)

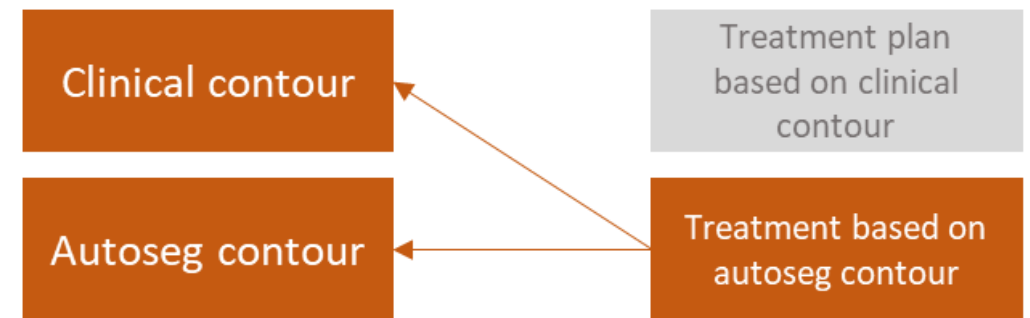
Problem : We do not have the treatment plan based on the auto-seg contour set

We use the DIA as a surrogate for the DIE

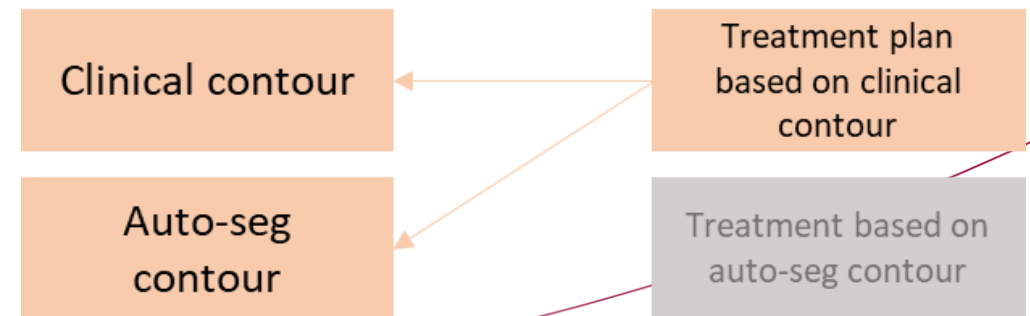
$$DIA(Gy) = D_{mean}(clinical) - D_{mean}(auto)$$

It quantifies the dosimetric difference introduced by contour discrepancies under a fixed treatment plan.

DIE = dosimetric impact from evaluation



Non-replanned (DIA)



Edits and their dosimetric effect during the years

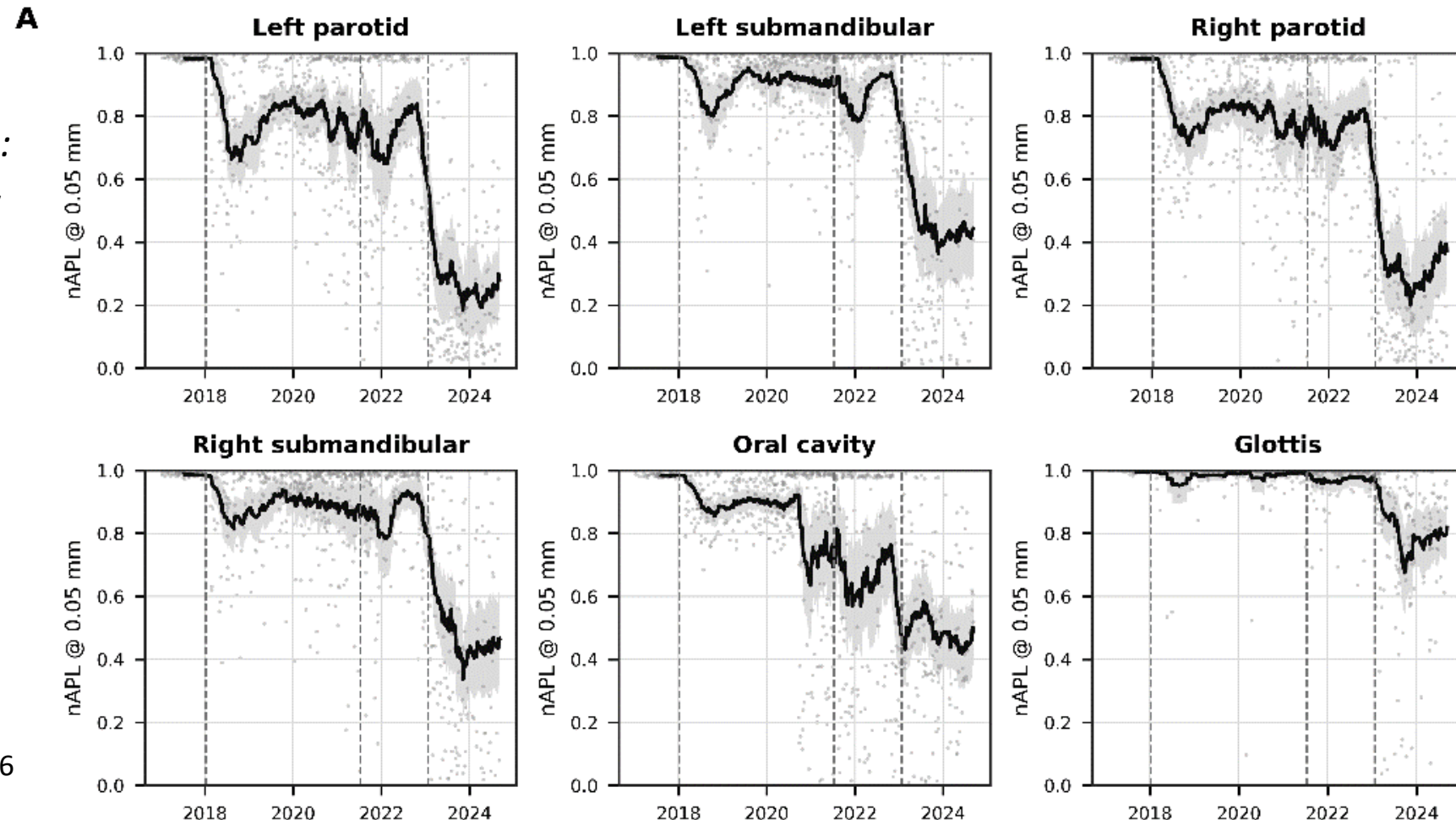
Geometric analysis → nAPL

nAPL (normalized added path length) : added contour length^[1] normalized by the clinical contour path length. A tolerance of 0.05mm was applied to ensure sensitivity to small contour edits^[2].

766 included patients.

[1] Vassen et al. Phys Imaging Radiat Oncol 2020

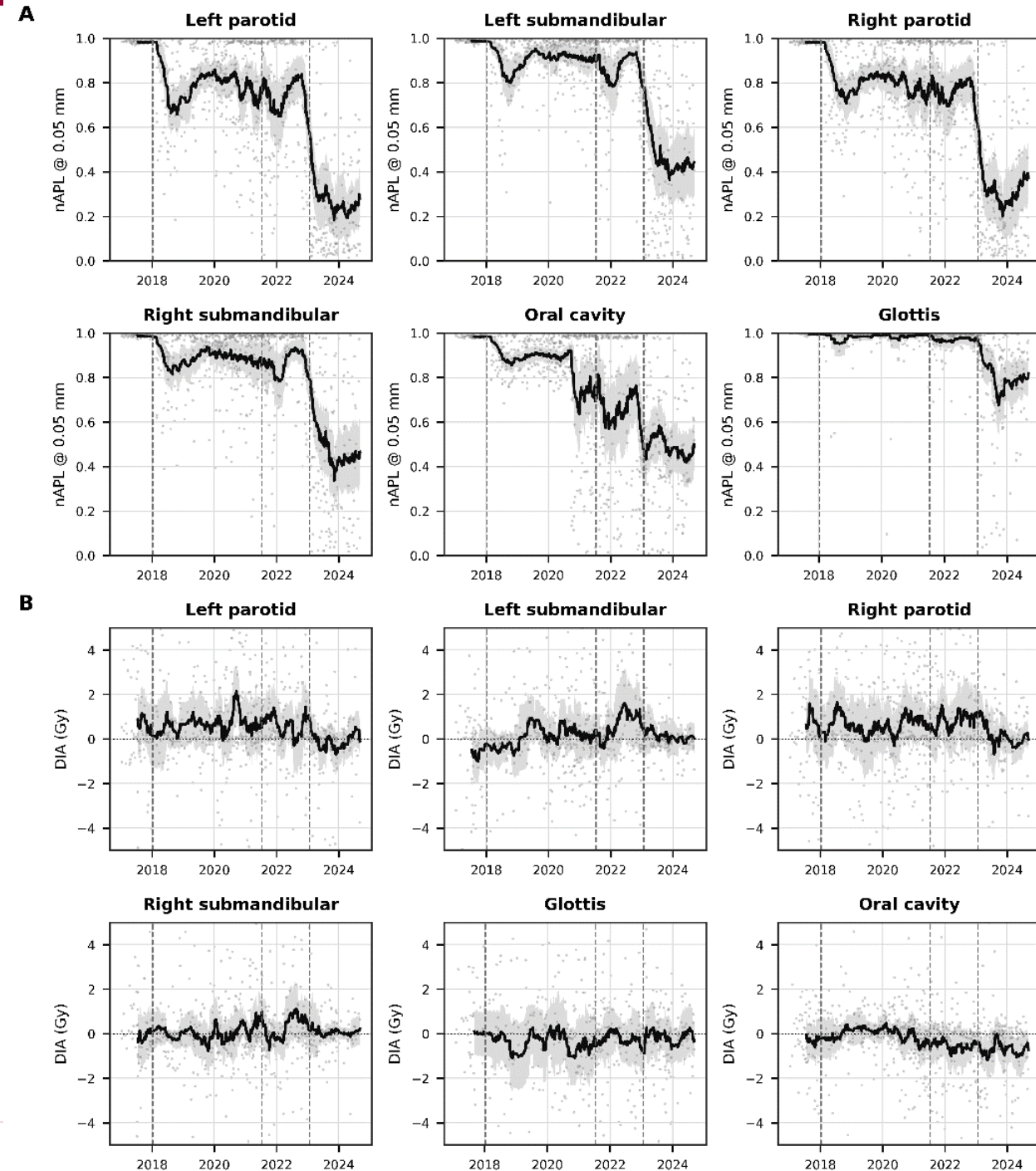
[2] Simões et al. Radiotherapy and Oncology 2026



Edits and their dosimetric effect during the years

Geometric (nAPL) vs dosimetric (DIA) analysis
766 included patients.

DIA (Gy) variability quite constant across years, regardless of change in auto-seg models.

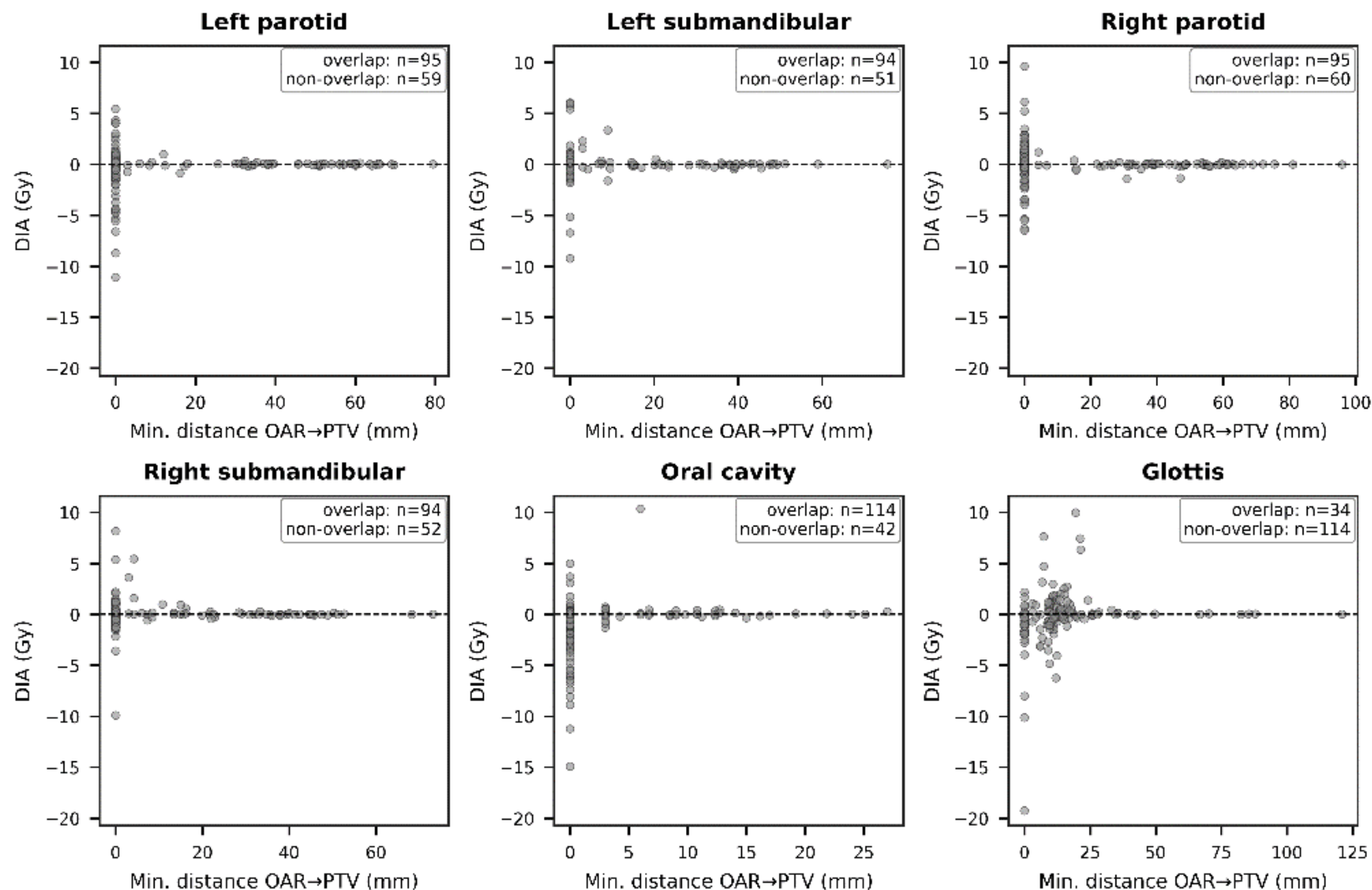


AIM (2): Determine the relationship between OOI-PTV distance and dosimetric impact, and translate it into editing recommendations

Relationship between OOI-PTV distance and dosimetric impact

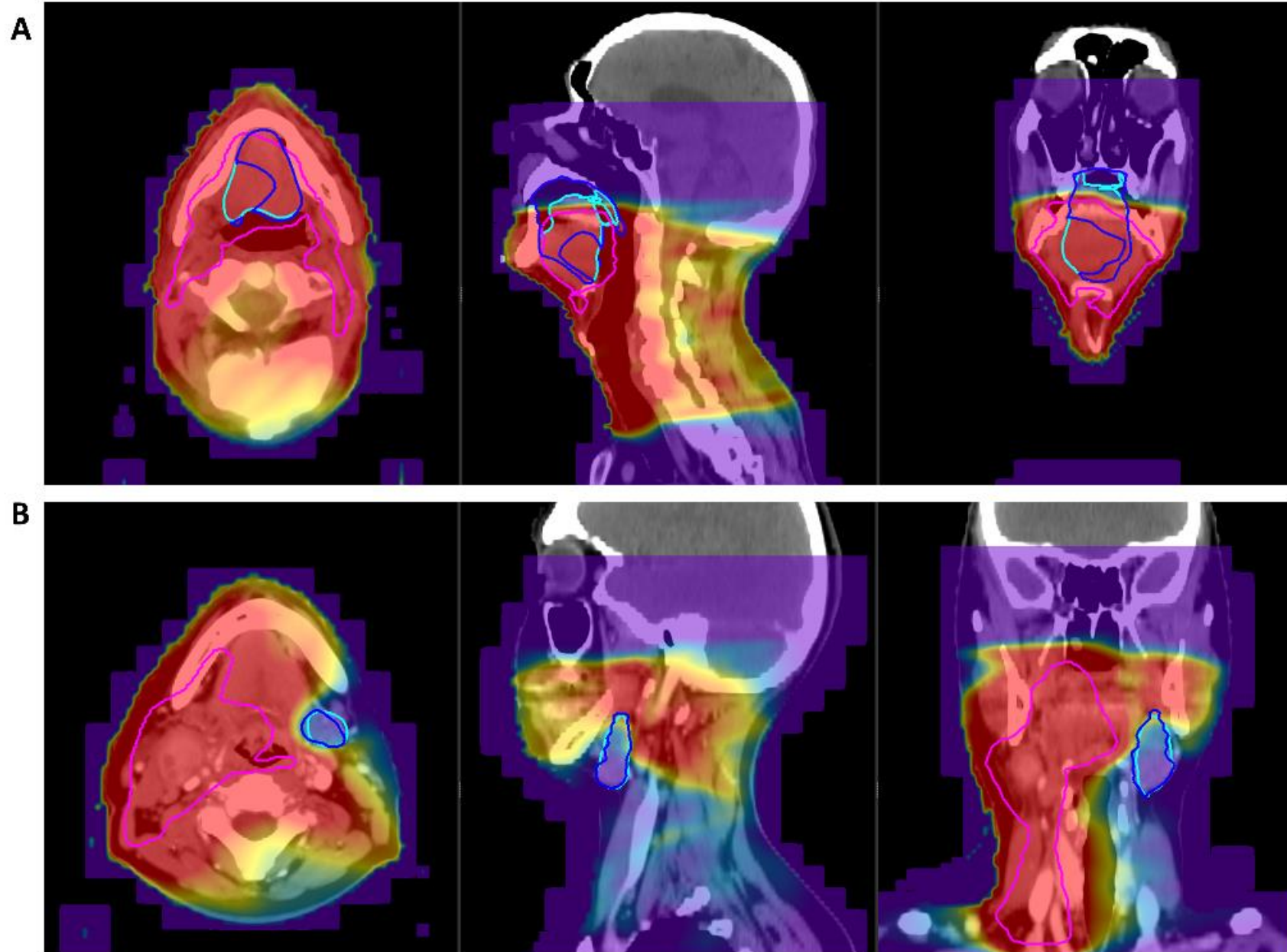
DIA was strongly associated with the minimum OOI-PTV distance, with larger variations observed for structures in closer proximity to the target.

Glottis: It exhibits broader variability in DIA → Relatively small size and anatomical complexity.



Example

Despite similar amounts of contour editing (nAPL = 0.705 and 0.753), an (A) overlapping oral cavity contour resulted in a large dosimetric difference (DIA = -14.9 Gy), whereas a (B) non-overlapping left submandibular gland resulted in a negligible dosimetric difference (DIA = 0.171 Gy).



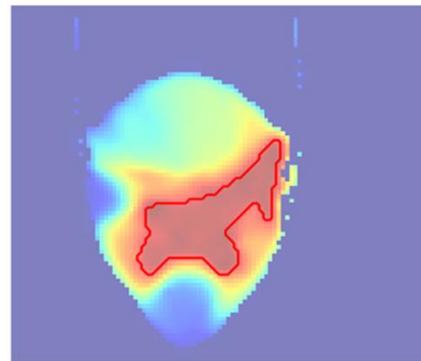
Translating OOI-PTV relationship into editing recs

Hybrid segmentations

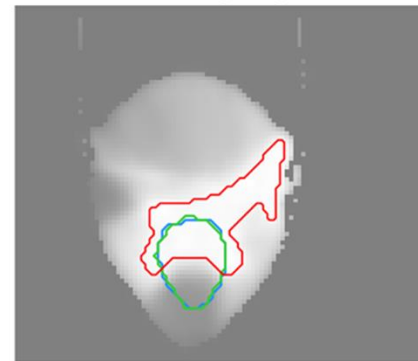
Hybrid segmentation reflects a clinical workflow in which contour edits are applied only for that region of the OOI located within a predefined distance to the PTV.

$$OOI (hybrid) = \underbrace{OOI \cap PTVd}_{\text{Clinical segmentation}} + \underbrace{OOI \cap \sim PTVd}_{\text{Auto-segmentation}}$$

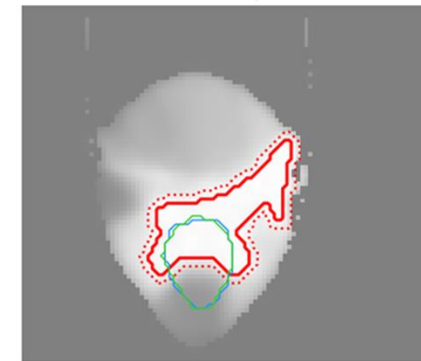
For $PTVd = [-3, 5]cm$ in 0.5 cm steps



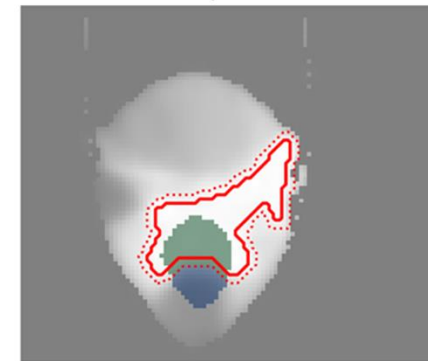
PTV + Dose



Blue = auto-seg
Green = clinical seg



PTV expanded



Hybrid seg

Translating OOI-PTV relationship into editing recs

Hybrid segmentations

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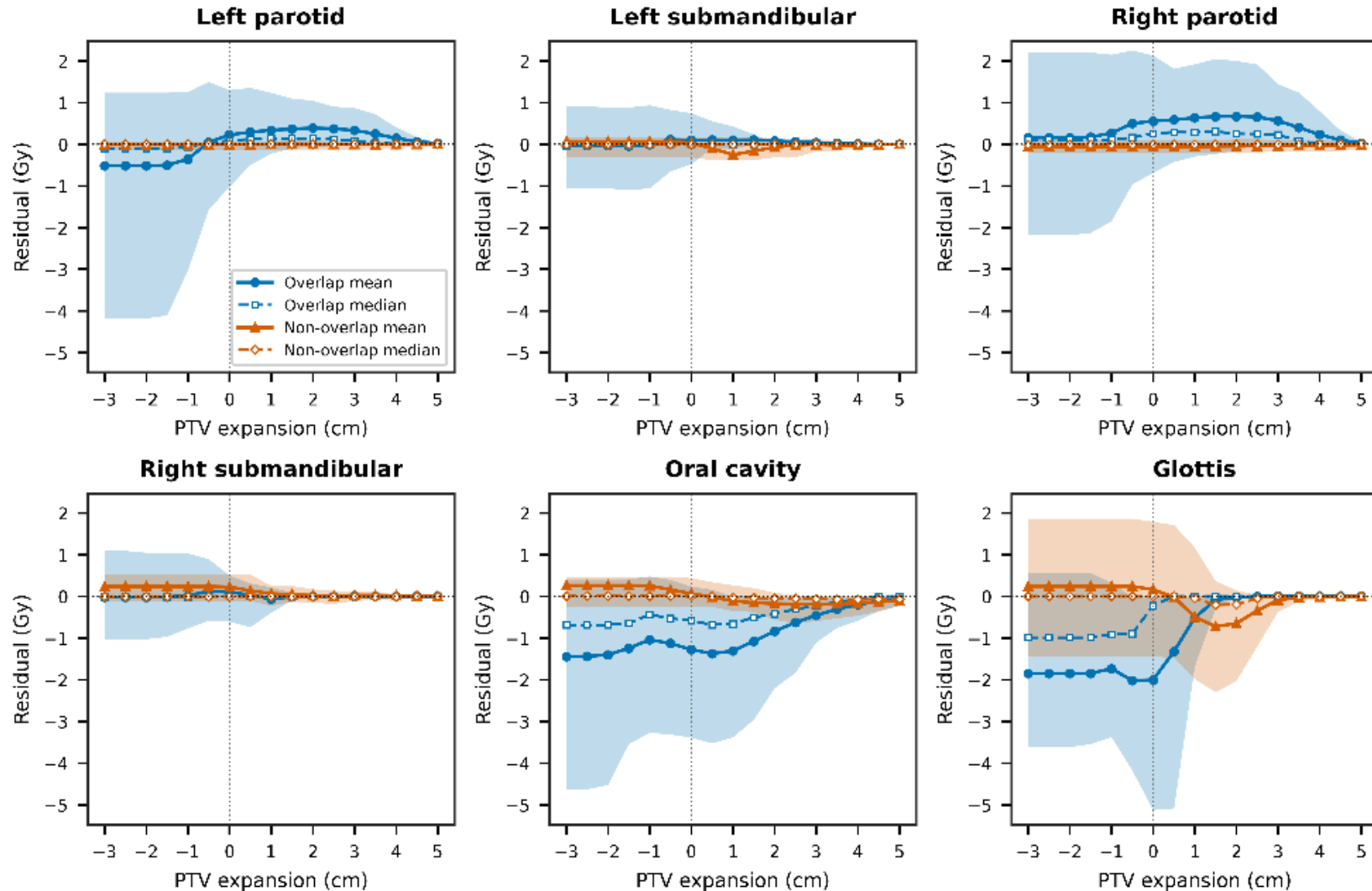
For $PTVd = [-3, 5]cm$ in 0.5 cm steps

To quantify the portion of the total dosimetric impact not captured when edits are restricted to a given spatial region around the PTV, the residual dosimetric difference was defined as

$$Residual = D_{mean}(clinical) - D_{mean}(hybrid)$$

Hybrid seg analysis and residual dose differences

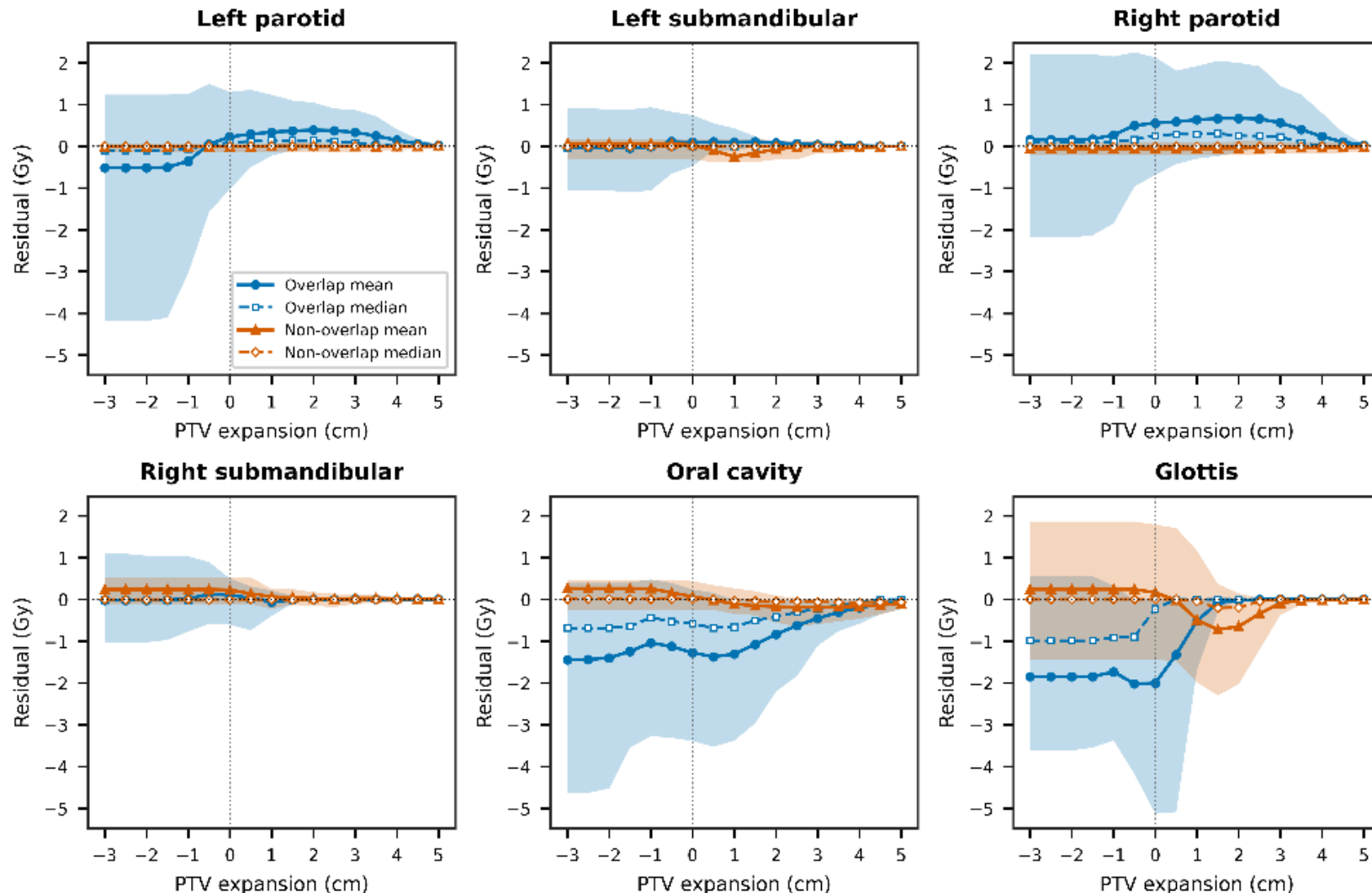
For **overlapping OOs**, restricting edits to within 2 cm of the PTV reduced residual differences to below 0.1 Gy for the submandibulars and glottis, and below 2.2 Gy for the parotids and oral cavity for 80% of patients.



Hybrid seg analysis and residual dose differences

For overlapping OOLs, restricting edits to within 2 cm of the PTV reduced residual differences to below 0.1 Gy for the submandibulars and glottis, and below 2.2 Gy for the parotids and oral cavity for 80% of patients.

Non-overlapping OOLs exhibited minimal residual differences even in the absence of editing, generally remaining below 0.5 Gy across all considered distances, indicating limited dosimetric benefit from contour corrections far from the PTV.



Conclusion

AIMS : (1) Assess the relationship between manual editing patterns and their dosimetric impact

Substantial variability in manual contour editing translated into only limited dosimetric impact for most OOI's, highlighting a disconnect between editing effort and clinical relevance.

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Substantial variability in manual contour editing translated into only limited dosimetric impact for most OOI, highlighting a disconnect between editing effort and clinical relevance.

AIMS : (2) determine the relationship between OOI-PTV distance and dosimetric impact and translate it into editing recommendations.

While occasional larger dose differences were observed, particularly for structures located within the PTV, the dosimetric relevance of contour edits was generally more closely associated with the spatial relationship between the OOI and the target rather than the extent of editing itself.

Conclusion

Editing recommendations

1. No routine editing for non-overlapping OOs (provided that the auto-segmentation has not failed).
2. Restricting edits of overlapping submandibular OOs to regions within 2cm of the PTV.

Collaborators

Rita Simões

Tomas Janssen

Casper Carbaat

H&N tumor group

THANKS!